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Bridging Industry and Academia: Co-Creating Interaction Design and Artificial Intelligence Courses

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Abstract. This paper explores the co-creation process between industry and academia in the development of courses within Interaction Design and Artificial Intelligence. Through a collaborative workshop with industry representatives and a survey targeting current students and alumni, we gathered insights into their needs, expectations, and priorities for future educational offerings in these fields. The co-creation process resulted in the development of several new course proposals, which were assessed by students and alumni coming from two bachelor programs. The findings present six courses with descriptions alongside their ratings. We also present qualitative feedback from students and alumni regarding their perceptions and anticipated value of these courses. The study highlights the importance of engaging industry stakeholders, students and researchers in curriculum development to ensure academic relevance and industry alignment. This work contributes to ongoing discussions on collaborative curriculum design in higher education, particularly in fields undergoing rapid change.

Keywords: Co-creation · workshop · survey · courses · Interaction Design · Artificial Intelligence

1 Introduction and Background

Educational practices and curricula development in Interaction Design (IxD) and Artificial Intelligence (AI) are individually well established, with each field benefiting from decades of experience and refinement. As AI becomes increasingly integrated into applications, its influence on IxD is growing (De Silva and Halloluwa, 2025), but there is still a gap in educational literature and curricula that address their intersection and the alignment with expectations and needs from students and the current market. Studies have discussed the value of interdisciplinary AI education, yet most initiatives remain either technically or design-oriented without clear directions of convergence (Kim et al., 2025; Kok et al., 2024). Reviews of curriculum development suggest limited examples

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where interaction-focused pedagogy is paired with AI learning outcomes (Ntunka & Maladzhi, 2024). Other works highlight the lack of integration between AI platforms and human-centred approaches in higher education (Azhar et al., 2024; Liu et al., 2024). Design students engaging with smart home technologies revealed that the existing design education does not adequately address current issues in automation and human-machine interdependence (Kuo & Xia, 2023).

As a further challenge, in the evolving landscape of IxD and AI education, academia and industry often find themselves at crossroads, each with different priorities and approaches (Windl, 2022; Wyatt & Piggott, 2019). Traditionally, academic institutions stress theoretical knowledge, encouraging students to investigate foundational ideas and participate in critical thinking, while the industry demands professionals who can apply knowledge to practical solutions that meet urgent market needs. IxD students, for example, may face difficulties articulating their professional roles and expected tasks in industry contexts (Sørum, 2017). A study further emphasizes the need to align AI curricula with evolving job market demands and suggests developing frameworks through closer collaboration between academia and industry (Li et al., 2019). Co-creation with industry can be one way to foster a more synergistic relationship between academia and industry, ensuring that course contents remain relevant and reflective of current industry practices (Laundon, 2023).

This paper reports on a co-creation effort related to the development of advanced courses for two new Master-level educational initiatives: a specialization in Human-Computer Interaction (HCI) and a Master program in Interactive AI Systems. The outcome of the study revealed a list of six courses that were proposed by industry representatives and assessed by student participants. The description of the courses along with their ratings from student surveys, provide a solid foundation for further development and accreditation of the proposed courses.

2 Methodology

The data collection for this study employed two methods: a workshop with companies and a survey with students and alumni. The two methods were applied in sequence, where the survey was used to assess the acceptance of the results from the workshop. By assessing the data from both perspectives, we intended to strengthen the trustworthiness of the research and allow for a richer interpretation of the results, promoting a more holistic and nuanced understanding of the findings through different lenses (Creswell and Creswell, 2022).

2.1 Workshop with Companies

Representatives from four different Swedish companies took part in the workshop, which lasted around three hours and took place within a Swedish university. The size of the companies ranged from small (few employees) to large (several hundreds), and they provide products and services within software design and development, education technologies, and forestry. Two of the participants had a strong background in IxD, which constituted their professional role within their respective companies, while the other two participants had leadership positions.

A visual presentation was created to introduce the central theme of the workshop and the five-step process to be followed, inspired by the Design Thinking approach (Lewrick, Link & Leifer, 2018): *problematisation*, *ideation*, *design*, *sharing* and *closing* (see Fig. 1). During the workshop introduction, participants were informed that the aim was to get their insights and opinions about potential courses on advanced level within the topics of IxD and Interactive AI, contributing to aligning educational offerings with industry needs.

In the *problematisation* phase we set out to understand the participants' needs and visions, using a set of interview cards designed with two categories of questions inspired by the SUE influence framework (2024): (1) current concerns in today's education (pains and comforts of actual professionals) and (2) tomorrow's education (gains and anxieties of new professionals). The *job-to-be-done* presented was to identify what skills and knowledge the industry needs from high-quality future professionals in AI and IxD.

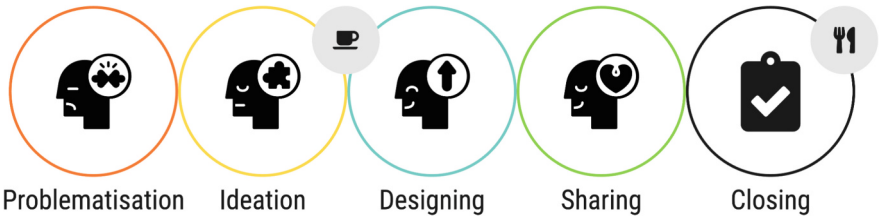


Fig. 1. The five workshop phases conducted with representatives from the industry.

During the *ideation* phase participants were instructed to use post-it notes to express all their ideas for courses and to write down as many ideas as possible. For each idea, they were asked to think about a possible scenario, i.e., expand on what each course would be like, what the main activities would be, the technology to be used, how to use it, etc.

These initial ideas were further expanded during the *designing* phase, where participants were arranged in groups and instructed to propose at least two courses by following a provided template. For each course, participants were requested to indicate how they would help future professionals to be more attractive in the market. The course template had fields such as name of the course, focus (AI/IxD), main skills, and content.[1]

During the *sharing* phase, participants presented the courses they had brainstormed. Each group had 10 min to present their ideas to the audience. Finally, in the *closing* phase, all attendees reflected on the workshop, the main takeaways, and the potential future work.

2.2 Survey with Students and Alumni

A survey comprising 15 questions was designed and distributed to current students and alumni from two bachelor programs at a Swedish university: Interaction Design (IxD) and Web Programming (WP). The IxD program focuses on delivering courses covering the topics of user involvement and understanding their needs, interface requirements,

[1] The survey and workshop materials can be found here: https://osf.io/t6g3f/files

prototyping and evaluation. With strong focus on qualitative methods, the program utilizes approaches such as Participatory Design, User-Centred Design, Design Thinking and Inclusive Design to design usable and accessible interfaces. The WP program focuses on programming and the aim is to create web applications for different platforms using methods that are common in the industry. The education focuses on code quality, security, and testing. Both programs are carried out in close collaboration with the business community.

The first seven questions were demographic and inquired whether the participants were current or former students, which program they were enrolled in, which mode (campus or distance), and whether they are currently employed. Then participants were presented with the six courses that were proposed during the workshop with companies (which are presented in Sect. 3, Results and Discussion). For each course, participants could read a concise overview of its objectives and the knowledge they would acquire. Participants were asked to rate each course in a 5-item Likert scale (*Not Interested* to *Very Interested*) and answer optional open-ended questions about their motivation for the given rating, the content they would like to see in the course, and any other comments. Participants were also encouraged to indicate their preference for the course to be offered on campus, distance or both. Finally, participants could also provide suggestions for other courses within related topics that we had not covered in the survey.

It took approximately 10 min to complete the survey, which was open for 14 days. It was hosted within the university platform and the dissemination was done using emails and Slack. Before starting the survey, the participants were informed about the purpose of the survey, the voluntary nature of participation, and ethical and confidentiality aspects regarding data collection and storing. In total, 61 respondents shared their opinions; 40 from the IxD program and 21 from the WP program.

3 Results and Discussion

3.1 Workshop Outcomes

From the first phase, the discussions revolved around various topics that could be represented into courses, such as AI, accessibility, tools and platforms for learning and prototyping, research and practical skills, etc. As we progressed through the workshop phases, participants were encouraged to focus and propose concrete courses, specifically providing the following: title of the course, focus of the course, skills it develops, content, and anything else they wished to add. Among several course ideas that were proposed and discussed, the following were the main outcomes of this phase:

- **Novel Interaction Methods using AI.** This course proposes a blend of traditional Human-Computer Interaction methods with AI, moving beyond basic aspects like "prompting". Specifically, the course should focus on building interfaces with AI-driven applications to enhance their UX. Additionally, aspects of requirements, prototyping, testing and stakeholder involvement, were suggested as content for the course. Finally, the course should also include mentorship elements that can help strengthen ties with companies.

- **Accessibility/Inclusive Design.** Being aware of the existing accessibility regulations and their increased relevance, participants highlighted the importance of developing and offering a course within accessibility and inclusive design. They explained that the course should focus on a broad spectrum of disabilities, but also include temporary impairment cases (e.g., injury). The course should teach students how to test websites and apps for accessibility issues, fix them, and understand user needs from an accessibility standpoint.
- **Process-based Understanding of an Organisation.** This course was proposed to offer students an understanding of an organisation and their role in it. Specifically, the course should cover aspects of managing and steer processes, and how one becomes an enabler within an organisation. The course should start by offering an overview of the overall organisation process and then moving closer to specific roles and processes.
- **AI Non-functional Requirements.** This course should focus on AI from the non-functional aspects, such as business principles of AI, security aspects, and GDPR as a framework to discuss ethical concerns and regulations. Data integration and transparency should also be covered.
- **Business Agility and AI.** This course should initially cover the fundamentals of understanding business agility, which later should be explored and evaluated in relation to AI and how it will affect it. Students should learn to work under simulated constraints that companies face. Additionally, they should comprehend how companies evaluate progress and outcomes.
- **Tools with AI.** Much of the discussion in the workshop was around AI and how we should teach about the fundamentals of AI. In relation to this, the focus of this course was to improve tools that are currently used by students in the education environment by leveraging recent advances in AI. Specifically, students would practice using tools to generate prototypes. (e.g., Figma, Adobe XD), analyse data (e.g., Microsoft Excel), and presenting (e.g., Microsoft PowerPoint) coupled with recent AI enhancements.

3.2 Survey Outcomes

The aim of the survey with current students and alumni was to gather feedback on the six courses that we identified as most relevant during the workshop, which we described in the previous section. The course that received the overall highest score was Accessibility/Inclusive Design with a mean value of 4.23 (out of 5), followed by Tools with AI (M: 4.07) and Novel Interaction Methods Using AI (3.80). Interestingly, these three courses received the highest rating in the same order by both, Interaction Design) and Web Programming (WP), respondents. This demonstrates strong consensus for these three courses regardless of the program. The means and 95% confidence intervals (CIs) for the ratings of the six courses are shown in Table 1, where the courses are ordered from highest to lowest in terms of perceived importance. The three highest-rated courses are further developed for integration into different study programs.

The open comments provided by some respondents offer further insights into their opinions regarding these courses. Their comments mainly discuss practical AI and IxD aspects, emphasizing the educational needs and desires of students. Students recognize the increasing importance of AI skills in the IxD field, as suggested by one respondent:

"I want to get benefits from AI tools in interaction design as much as possible to save time and effort. It is really important to have the ability to adapt AI tools in our courses."

Table 1. The list of courses with ratings offered by participants.

Course	Mean and 95% CI
Accessibility/Inclusive Design	4.23 ± 0.276
Tools with AI	4.07 ± 0.303
Novel Interaction Methods Using AI	3.80 ± 0.326
Process-based Understanding of an Organization	3.36 ± 0.346
AI Non-functional Requirements	3.33 ± 0.292
Business Agility and AI	3.15 ± 0.329

Students expressed desires to integrate these technologies into practical applications within design, with similar results found in other studies (Cavallin, 2024; Lively and Hutson, 2024). Specifically, they emphasize the need for courses that focus on hands-on projects and case studies, ensuring students can apply theoretical knowledge effectively. These courses should ideally bridge the gap between theory and real-world applications, as one respondent stated:

"I would also like to learn practical methods for integrating AI into business processes to better adapt to evolving customer needs, market trends, and operational challenges."

and another stating:

"Preferably a lot of focus on the practical use of AI."

Considering these expressed needs, students suggest courses focused on AI usage in pitching ideas portfolio development and professional presentation skills. Interestingly, such needs point to a pedagogical shift that prioritizes practical AI fluency and iterative skill development within curricular design. To this end, partnerships with industry could provide valuable resources for further course development.

Reflecting on the difference between IxD and WP respondents, the comments show that IxD students showed a deeper preference for AI integration into design processes, such as user experience enhancement, prototype development, and creative process documentation. Additionally, they emphasized the needs for learning about tools and methodologies within design. WP students, on the other hand, concentrated more on the practical applications like efficiency gains, using AI to streamline workflows for time-saving purposes, portfolio development, and enhancing presentation skills. These comments were in relation to courses: Novel interaction methods using AI, AI-nonfunctional requirements, and Tools with AI. One WP student, however, commented:

"Since generative AI for public use is relatively new, it feels uncertain and risky to develop a course about this, as our use of the tool will probably develop drastically in the coming years."

This insight raises concerns that it could require significant time to keep the course updated, and the skills learned might quickly become irrelevant.

IxD and WP respondents, showed similar inclinations towards acknowledging the importance of accessibility principles, with IxD respondents expressing the need to integrate these into curricula for inclusivity, whereas WP respondents focused on understanding and meeting legal accessibility aspects in their solutions with emphasis on technical aspects, as one respondent stated:

"...teach accessibility in web development and not just UX (e.g., ARIA, navigation with keyboard/without mouse, etc.)".

Similar findings are reported in studies by Lazar et al. (2019) and Keith and Whitney (2011). Both cohorts recommended the accessibility course should cover POUR[2] (Perceivable, Operable, Understandable, and Robust) principles and WCAG[3] (Web Content Accessibility Guidelines) guidelines, as one respondent stated:

"Being able to understand the meaning of POUR in a deeper way would have been good to get a better understanding of why WCAG has developed its guidelines."

and another stating that:

"Many companies are now forced to follow WCAG laws so it sounds like a very good course."

Respondents' feedback—especially their desire to understand POUR and WCAG not just in theory but through implementation—suggests that curriculum development should not only acknowledge accessibility as a principle but also as a practical, legislatively grounded, skill set. Furthermore, Burgstahler and Cauce (2020) highlight that hands-on learning and engagement with legal frameworks like WCAG foster deeper understanding and long-term competency among students.

Respondents from two programs also shared a common sentiment regarding business-oriented courses, finding them too broad and not directly relevant to their roles. In summary, IxD students appeared more focused on AI integration into design processes and methodologies, whereas WP students emphasized practical applications in efficiency and professional skills.

4 Conclusion

This study reports on the co-creation process of advanced courses within Interaction Design and AI. The results from a workshop with industry representatives and a survey with current students and alumni proposed six courses that represent a blend of IxD and AI. These findings echo broader trends in design and computing education, where there is a growing emphasis on equipping students with adaptive, future-facing skills in AI integration.

Three courses, *Novel Interaction Methods*, *Inclusive Design*, and *Tools with AI*, are being further developed. The proposed content of the first two courses aligns well with the HCI field and are planned to be offered as part of an HCI specialization track within the Master's program in Informatics.[4] The decision to include these two courses within the HCI specialization, is pedagogically grounded in the need to address emerging domains such as inclusive design and AI-enhanced interaction that extend beyond traditional UX or HCI content. These courses provide focused depth and research-oriented competencies aligned with evolving professional and academic demands. The delivery of these courses supports industry collaboration and a balanced integration between foundational HCI principles and novel application areas.

[2] https://ialabs.ie/understanding-the-pour-principles-of-accessibility/

[3] https://www.w3.org/TR/WCAG22/

[4] https://lnu.se/en/programme/informatics-master-programme/vaxjo-international-autumn/

The first course (*Novel Interaction Methods*) will also be offered as an elective within the Master's program in Interactive AI Systems.[5] The third course (*Tools with AI*) is being considered to be offered in one of the existing bachelor programs.

The study is rather limited in terms of the number of companies involved and geographical scope with participants from two programs mainly coming from the region of south Sweden. With such limitations in mind, the study can be replicated to reach a wider scope and diversity of participants. In addition, the concrete near future plan is the three highest rated courses to be further developed, accredited, and offered within the respective advanced programs.

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